

■ PE better   
 ■ no difference   
 ■ Standard better   
    outlined: 95% CI excludes 0

Rotated shifted Rastrigin

| Dimension $d$ | 1/2                     | 2/3   | 5/6   | 1            | 7/6   | 4/3   | 3/2   |
|---------------|-------------------------|-------|-------|--------------|-------|-------|-------|
| 2             | <b>+0.81</b>            | +0.61 | -0.10 | -0.30        | +0.39 | +0.20 | +0.28 |
| 5             | +0.03                   | -0.13 | -0.22 | <b>+0.15</b> | -0.57 | -0.48 | +0.00 |
| 10            | +0.59                   | +0.69 | -0.60 | -0.30        | +0.32 | +0.01 | -0.15 |
| 20            | +3.40                   | +0.99 | -0.09 | <b>-4.03</b> | -2.60 | -3.59 | -1.85 |
| 30            | +1.89                   | +1.95 | +2.84 | +2.57        | +2.74 | +0.44 | +1.48 |
|               | 1/2                     | 2/3   | 5/6   | 1            | 7/6   | 4/3   | 3/2   |
|               | Decay exponent $\alpha$ |       |       |              |       |       |       |

Rosenbrock

| Dimension $d$ | 1/2                     | 2/3           | 5/6           | 1             | 7/6    | 4/3    | 3/2    |
|---------------|-------------------------|---------------|---------------|---------------|--------|--------|--------|
| 2             | <b>-0.007</b>           | -0.002        | <b>-0.010</b> | <b>-0.009</b> | -0.003 | -0.003 | -0.003 |
| 5             | -0.027                  | <b>-0.018</b> | +0.000        | -0.002        | +0.005 | -0.003 | -0.006 |
| 10            | +0.011                  | <b>+0.030</b> | +0.010        | -0.001        | -0.013 | -0.000 | -0.000 |
| 20            | <b>+0.153</b>           | <b>+0.151</b> | <b>+0.150</b> | +0.132        | +0.115 | +0.113 | +0.096 |
| 30            | +0.241                  | <b>+0.222</b> | +0.257        | +0.235        | +0.200 | +0.096 | +0.073 |
|               | 1/2                     | 2/3           | 5/6           | 1             | 7/6    | 4/3    | 3/2    |
|               | Decay exponent $\alpha$ |               |               |               |        |        |        |

Shifted Ackley

| Dimension $d$ | 1/2                     | 2/3          | 5/6   | 1            | 7/6   | 4/3   | 3/2          |
|---------------|-------------------------|--------------|-------|--------------|-------|-------|--------------|
| 2             | <b>+0.04</b>            | +0.01        | -0.50 | -0.02        | -0.49 | -0.51 | -0.01        |
| 5             | +0.14                   | <b>-0.66</b> | +0.25 | <b>-1.29</b> | -0.27 | -0.11 | +0.19        |
| 10            | +0.56                   | +0.10        | +0.16 | +0.25        | +0.47 | +0.49 | +0.04        |
| 20            | +0.09                   | +0.36        | -0.05 | +0.36        | -0.24 | -0.05 | +0.35        |
| 30            | +1.30                   | <b>+1.18</b> | +1.40 | <b>+1.59</b> | +0.55 | +0.90 | <b>+1.16</b> |
|               | 1/2                     | 2/3          | 5/6   | 1            | 7/6   | 4/3   | 3/2          |
|               | Decay exponent $\alpha$ |              |       |              |       |       |              |

Cells show PE improvement over Standard for Decay Uniform under GP-UCB, 400 BO steps, 5 paired seeds. Colour is scaled symmetrically about zero, independently per benchmark.